Quine McClusky Method(Tabulation Method0

TABULAR METHOD FOR MINIMIZATION

- K-map method of simplification is convenient as long as the number of variables does not exceed five or six.
- As the number of variables increases it is difficult to make judgement about which combinations form the minimum expression. (Complexity of **K-map** simplification process increases with the increase in the number of variables.(The minimum expression obtained might not be unique.))
- K-map simplification is manual technique and simplification process is heavily depends on the human abilities.
- To meet this needs, In 1950s W.V Quine and E.J McCluskey developed an exact tabular method to simplify the Boolean expression. This method is called the Quine-McCluskey, or tabular method.

Algorithm for generating minimum prime implicants using Quine-McCluskey/ Tabular method:

I^{st} step

I

1. List all minterms in the binary form.
2. Arrange the minterms according to number of 1s.
3. Compare each binary number with every term in the adjacent next higher category and if they differ only by one position, put a tick mark and copy the term in the next column with ‘_ or X’ in the position that they differed.
4. Apply the same process described in step 3 for the resultant column and continue these cycles until a single pass through cycle yields no further elimination of literals.
5. List all prime implicants.

II

6. Select the minimum number of prime implicants which must cover all the minterms.

└ PI table

Generate the prime implicants for the function

$$f(x1,x2,x3,x4) = \sum m(0,4,8,10,11,12,13,15)$$

0000, 0100, 1000, 1010, 1011, 1100, 1101, 1111

Step I — Generate PI

Col 1	$x_1x_2x_3x_4$	Col 2	$x_1x_2x_3x_4$	Col 3	$x_1x_2x_3x_4$
Grp 0: 0	0000 ✓	(0,4)	0_00 ✓	(0,4,8,12)	--00
Grp 1: 4	0100 ✓	(0,8)	_000 ✓	(0,8,4,12)	--00
8	1000 ✓	(4,12)	_100 ✓		
Grp 2: 10	1010 ✓	(8,10)	10_0		
12	1100 ✓	(8,12)	1_00 ✓		
Grp 3: 11	1011 ✓	(10,11)	101_		
13	1101 ✓	(12,13)	110_		
Grp 4: 15	1111 ✓	(11,15)	1_11		
		(13,15)	11_1		

Prime implicants:

1. $x_1\bar{x}_2\bar{x}_4$
2. $x_1\bar{x}_2x_3$
3. $x_1x_2\bar{x}_3$
4. $x_1x_3x_4$
5. $x_1x_2x_4$
6. $\bar{x}_3\bar{x}_4$

Generate the prime implicants for the function

$$f(x1,x2,x3,x4) = \sum m(0,4,8,10,11,12,13,15)$$

List 1

0	0 0 0 0	✓
4	0 1 0 0	✓
8	1 0 0 0	✓
10	1 0 1 0	✓
12	1 1 0 0	✓
11	1 0 1 1	✓
13	1 1 0 1	✓
15	1 1 1 1	✓

List 2

0,4	0 x 0 0	✓
0,8	x 0 0 0	✓
8,10	1 0 x 0	
4,12	x 1 0 0	✓
8,12	1 x 0 0	✓
10,11	1 0 1 x	
12,13	1 1 0 x	
11,15	1 x 1 1	
13,15	1 1 x 1	

List 3

0,4,8,12	x x 0 0

$$P = \{10x0,\ 101x,\ 110x,\ 1x11,\ 11x1,\ xx00\}$$
$$= \{p_1, p_2, p_3, p_4, p_5, p_6\}$$

DETERMINATION OF A MINIMUM COVER

Prime implicant	0	4	8	10	11	12	13	15
	Minterm							
p_1 = 1 0 x 0			✓	✓				
p_2 = 1 0 1 x				✓	✓			
p_3 = 1 1 0 x						✓	✓	
p_4 = 1 x 1 1					✓			✓
p_5 = 1 1 x 1							✓	✓
p_6 = x x 0 0	✓	✓	✓			✓		

(a) Initial prime implicant cover table

II DETERMINATION OF A MINIMUM COVER

PI ↓ / Minterms →	0	4	8	10	11	12	13	15
① (8, 10)			X	X				
② (10, 11)				X	X			
③ (12, 13)						X	X	
④ (11, 15)					X			X
⑤ (13, 15)							X	X
⑥ (0, 4, 8, 12) *	X	X	X			X		

$\bar{x}_3\bar{x}_4$

$= \bar{x}_3\bar{x}_4 +$

single check mark in some column of the cover table, then the prime implicant that covers the minterm of this column is *essential* and it must be included in the final cover. Such is the case with p_6, which is the only prime implicant that covers minterms 0 and 4. The next step is to remove the row(s) corresponding to the essential prime implicants and the column(s) covered by them. Hence we remove p_6 and columns 0, 4, 8, and 12, which leads to the table in figure b.

Prime implicant	Minterm 10	11	13	15
(8,10) ~~p_1~~	✓			
(10,11) p_2	✓	✓		
(12,13) ~~p_3~~			✓	
(11,15) p_4		✓		✓
(13,15) p_5			✓	✓

(b) After the removal of essential prime implicants

$= p_1$

$= \bar{x}_3\bar{x}_4 + \,?$

✓ p_5 is dominant row over p_3 X

✓ p_2 is dominant row over p_1 X

Now, we can use the concept of *row dominance* to reduce the cover table. Observe that p_1 covers only minterm 10 while p_2 covers both 10 and 11. We say that p_2 *dominates* p_1. Since the cost of p_2 is the same as the cost of p_1, it is prudent to choose p_2 rather than p_1, so we will remove p_1 from the table. Similarly, p_5 dominates p_3, hence we will remove p_3 from the table. Thus, we obtain the table in Figure 4.37*c*. This table indicates that we must choose p_2 to cover minterm 10 and p_5 to cover minterm 13, which also takes care of covering minterms 11 and 15. Therefore, the final cover is

$$C = \{p_2, p_5, p_6\}$$
$$= \{101x, 11x1, xx00\}$$

$$= \bar{x}_3\bar{x}_4 + x_1\bar{x}_2x_3 + x_1x_2x_4$$

Prime implicant	Minterm 10	11	13	15
p_2	✓	✓		
p_4		✓		✓
p_5			✓	✓

(c) After the removal of dominated rows

which means that the minimum-cost implementation of the function is

$$f = x_1\bar{x}_2x_3 + x_1x_2x_4 + \bar{x}_3\bar{x}_4$$

(3) Use the tabular method to determine the minimum-cost SoP expression for the function f(x1,x2,x3,x4) = $\sum$m (4,6,8,10,11,12,15) + D(3,5,7,9)

List 1

4	0 1 0 0 √
8	1 0 0 0 √
3	0 0 1 1√
5	0 1 0 1 √
6	0 1 1 0 √
9	1 0 0 1 √
10	1 0 1 0 √
12	1 1 0 0 √
7	0 1 1 1 √
11	1 0 1 1 √
15	1 1 1 1 √

List 2

4,5	0 1 0 x √
4,6	0 1 x 0 √
4,12	x 1 0 0
8,9	1 0 0 x √
8,10	1 0 x 0 √
8,12	1 x 0 0
3,7	0 x 1 1 √
3,11	x 0 1 1 √
5,7	0 1 x 1 √
6,7	0 1 1 x √
9,11	1 0 x 1 √
10,11	1 0 1 x √
7,15	x 1 1 1 √
11,15	1 x 1 1 √

List 3

4,5,6,7	0 1 x x
4,6,5,7	0 1 x x
8,9,10,11	1 0 x x
8,10,9,11	1 0 x x
3,7,11,15	x x 1 1
3,11,7,15	x x 1 1

The prime implicants for this function are:

P = {x100, 1x00, 01xx, 01xx, 10xx, 10xx, xx11, xx11}

P = {x100, 1x00, 01xx, 10xx, xx11}

= { p1,p2,p3,p4,p5}

	Prime Implicant		Minterm 4	6	8	10	11	12	15
	P1	x 1 0 0	√					√	
3	P2	1 x 0 0			√			√	
1	P3	0 1 x x	√	√					
2	P4	1 0 x x			√	√	√		
4	P5	x x 1 1					√		√

$$f(x1,x2,x3,x4) = x1'x2 + x1x2' + x1x3'x4' + x3x4$$

	Prime Implicant		4	6	8	10	11	12	15
			Minterm						
	P1	x 1 0 0	√					√	
3	P2	1 x 0 0			√			√	
1	P3	0 1 x x	√	√					
2	P4	1 0 x x			√	√	√		
4	P5	x x 1 1					√		√

f = P3 + P4 + P5 + P1

or

P3 + P4 + P5 + P2

$$f(x1,x2,x3,x4) = x1'x2 + x1x2' + x1x3'x4' + x3x4$$

(4) $f(x_1, \ldots, x_4) = \sum m(0, 2, 5, 6, 7, 8, 9, 13) + D(1, 12, 15)$

(4) $f(x_1, \ldots, x_4) = \sum m(0, 2, 5, 6, 7, 8, 9, 13) + D(1, 12, 15)$

List 1

0	0 0 0 0	✓
1	0 0 0 1	✓
2	0 0 1 0	✓
8	1 0 0 0	✓
5	0 1 0 1	✓
6	0 1 1 0	✓
9	1 0 0 1	✓
12	1 1 0 0	✓
7	0 1 1 1	✓
13	1 1 0 1	✓
15	1 1 1 1	✓

List 2

0,1	0 0 0 x	✓
0,2	0 0 x 0	
0,8	x 0 0 0	✓
1,5	0 x 0 1	✓
2,6	0 x 1 0	
1,9	x 0 0 1	✓
8,9	1 0 0 x	✓
8,12	1 x 0 0	✓
5,7	0 1 x 1	✓
6,7	0 1 1 x	
5,13	x 1 0 1	✓
9,13	1 x 0 1	✓
12,13	1 1 0 x	✓
7,15	x 1 1 1	✓
13,15	1 1 x 1	✓

List 3

0,1,8,9	x 0 0 x
1,5,9,13	x x 0 1
8,9,12,13	1 x 0 x
5,7,13,15	x 1 x 1

$P = \{00x0, 0x10, 011x, x00x, xx01, 1x0x, x1x1\}$

$= \{p_1, p_2, p_3, p_4, p_5, p_6, p_7\}$

(0,2) (2,6) (6,7) (0,1,8,9) (1,5,9,13) (8,9,12,13) (5,7,13,15)

Prime implicant	Minterm 0	2	5	6	7	8	9	13
p_1 = 0 0 x 0	✓	✓						
p_2 = 0 x 1 0		✓		✓				
p_3 = 0 1 1 x				✓	✓			
p_4 = x 0 0 x	✓					✓	✓	
p_5 = x x 0 1			✓				✓	✓
p_6 = 1 x 0 x						✓	✓	✓
p_7 = x 1 x 1			✓		✓			✓

(a) Initial prime implicant cover table

Prime implicant	Minterm 0	2	5	6	7	8	9	13
p_1 = 0 0 x 0	✓	✓						
p_2 = 0 x 1 0		✓		✓				
p_3 = 0 1 1 x				✓	✓			
p_4 = x 0 0 x	✓					✓	✓	
p_5 = x x 0 1			✓				✓	✓
p_6 = 1 x 0 x						✓	✓	✓
p_7 = x 1 x 1			✓		✓			✓

col 9 is dom over col 8

col 13 is dom over col 5

(a) Initial prime implicant cover table

Prime implicant	Minterm 0	2	5	6	7	8
p_1 = 0 0 x 0	√	√				
p_2 = 0 x 1 0		√		√		
p_3 = 0 1 1 x				√	√	
p_4 = x 0 0 x	√					√
~~p_5 = x x 0 1~~			√			
~~p_6 = 1 x 0 x~~						√
p_7 = x 1 x 1			√		√	

(b) After the removal of columns 9 and 13

row P7 is dom over P5
row P4 is dom over P6
col 7 is dom over col 5
col 0 is dom on col

		2	5	6	8
(3)	P1	X			
(3)	P2	X		X	
2	P3			X	
2	P4				X
2	P7		X		

$f = P2 + P4 + P7$

Prime implicant	Minterm 0	2	5	6	7	8
p_1	✓	✓				
p_2		✓		✓		
p_3				✓	✓	
p_4	✓					✓
p_7			✓		✓	

(c) After the removal of rows p_5 and p_6

Prime implicant	Minterm 2	6
p_1	✓	
p_2	✓	✓
p_3		✓

(d) After including p_4 and p_7 in the cover

Prime implicant	Minterm 0	2	5	6	7	8
p_1	✓	✓				
p_2		✓		✓		
p_3				✓	✓	
p_4	✓					✓
p_7			✓		✓	

(c) After the removal of rows p_5 and p_6

Prime implicant	Minterm 2	6
p_1	✓	
p_2	✓	✓
p_3		✓

(d) After including p_4 and p_7 in the cover

After removing columns 9 and 13, we obtain the reduced table in Figure 4.39*b*. In this table row p_4 dominates p_6 and row p_7 dominates p_5. This means that p_5 and p_6 can be removed, giving the table in Figure 4.39*c*. Now, p_4 and p_7 are essential to cover minterms 8 and 5, respectively. Thus, the table in Figure 4.39*d* is obtained, from which it is obvious that p_2 covers the remaining minterms 2 and 6. Note that row p_2 dominates both rows p_1 and p_3.

The final cover is

$$
\begin{aligned}
C &= \{p_2, p_4, p_7\} \\
&= \{0x10,\ x00x,\ x1x1\}
\end{aligned}
$$

and the function is implemented as

$$f = \bar{x}_1 x_3 \bar{x}_4 + \bar{x}_2 \bar{x}_3 + x_2 x_4$$

5. f(w,x,y,z) = ∑m(0,1,3,4,7,11,13,15) + ∑d(9,12,14)

	0	1	3	4	7	11	13	15
~~P1~~ (0,1) ✓	x	x						
P2 (0,4) ✓	x			x				
~~P3~~ (4,12)				x				
P4 (1,3,9,11) ✓		x	x			x		
P5 (3,7,11,15)			x		x	x		x
P6 (9,11,13,15) ✓						x	x	x
~~P7~~ (12,13,14,15)							x	x

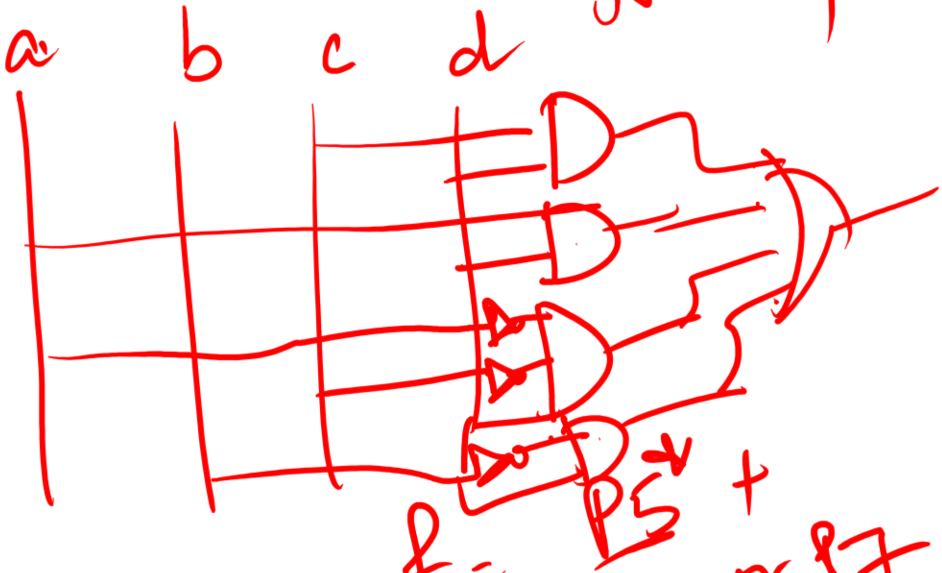
P1 is dom over P4

P6 or P7

f = P5 + – –

col 0 is dominated
wrt col 4

		0	1	4	13
(0,1)	P1	*	*		
(0,4)	P2	*		*	
(1,3,9,11)	P4		*		
(9,11,13,15)	P6				*



f = P5 + P6 or P7 + P2 + P4

$f = cd + \acute{a}d + \bar{a}\bar{c}d + \bar{b}d$

or

$cd + ab + \bar{a}\bar{c}d + \bar{b}d$

a → $\bar{a}$ complement

True

23 | 8 gates + 12 + 3